

## GRAND TRAVERSE LIGHT, MI, USA

WMO: 997731

Lat: 45.2106N

Lon: 85.5503W

Elev: 184

StdP: 99.13

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -15.7      | -13.4 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 18.8                  | -1.6 | 17.1 | -5.0 | 6.3                   | 280  | 0.626 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 8             | 6.9                    | 27.8            | N/A  | 26.3 | N/A  | 25.0 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 4.7                  | 230  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |     |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|-----|-----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |     |     |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |     |     |
| (a)               | (b)  | (c)  | (d)                        | (e) | (f)                | (g) | (h)                                                | (i) | (j)        | (k) | (l)        | (m) | (n)        | (o) | (p) | (q) |
| (4) 15.7          | 14.1 | 12.3 | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A | N/A | N/A |
| (5)               |      |      | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A | N/A | N/A |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 7.9    | -4.5  | -5.0  | -1.0  | 4.6   | 10.5  | 16.1  | 20.1  | 20.6  | 17.1  | 10.7  | 4.5   | -0.2  |
|                                                                        | DBStd       | 9.83   | 4.61  | 4.90  | 5.45  | 4.07  | 4.41  | 3.81  | 2.78  | 2.46  | 3.41  | 3.97  | 4.27  | 3.78  |
|                                                                        | HDD10.0     | 1957   | 451   | 421   | 344   | 170   | 45    | 1     | 0     | 0     | 0     | 37    | 172   | 317   |
|                                                                        | HDD18.3     | 4026   | 709   | 654   | 599   | 411   | 248   | 89    | 15    | 7     | 64    | 239   | 416   | 575   |
|                                                                        | CDD10.0     | 1174   | 0     | 0     | 3     | 9     | 61    | 183   | 313   | 328   | 212   | 60    | 6     | 0     |
|                                                                        | CDD18.3     | 201    | 0     | 0     | 0     | 0     | 5     | 21    | 69    | 76    | 26    | 3     | 0     | 0     |
|                                                                        | CDH23.3     | 713    | 0     | 0     | 1     | 1     | 40    | 88    | 253   | 238   | 83    | 10    | 0     | 0     |
|                                                                        | CDH26.7     | 101    | 0     | 0     | 0     | 0     | 8     | 19    | 30    | 32    | 11    | 1     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 5.2    | 6.9   | 6.2   | 5.0   | 4.8   | 3.9   | 3.4   | 4.1   | 4.5   | 4.9   | 5.9   | 6.0   | 6.4   |
| (15) Precipitation                                                     | PrecAvg     | 794    | 54    | 37    | 46    | 66    | 76    | 72    | 74    | 76    | 88    | 80    | 68    | 55    |
|                                                                        | PrecMax     | 943    | 105   | 73    | 112   | 116   | 140   | 187   | 169   | 129   | 167   | 177   | 133   | 132   |
|                                                                        | PrecMin     | 616    | 19    | 14    | 12    | 14    | 32    | 14    | 29    | 23    | 18    | 28    | 30    | 12    |
|                                                                        | PrecStd     | 78     | 24    | 15    | 26    | 26    | 28    | 39    | 36    | 31    | 39    | 41    | 28    | 25    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 8.2   | 6.9   | 19.3  | 22.6  | 28.1  | 29.7  | 29.8  | 29.6  | 28.1  | 24.7  | 17.8  | 12.2  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 2%          | DB     | 5.7   | 3.7   | 12.7  | 17.9  | 23.9  | 25.9  | 27.2  | 27.4  | 25.6  | 21.2  | 13.9  | 7.5   |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5%          | DB     | 3.6   | 2.2   | 8.5   | 14.2  | 20.6  | 23.6  | 25.7  | 25.8  | 23.8  | 18.6  | 11.7  | 6.1   |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 10%         | DB     | 2.1   | 1.2   | 5.7   | 10.8  | 17.8  | 21.7  | 24.7  | 24.6  | 22.2  | 16.2  | 10.1  | 5.0   |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 4.6   | 5.3   | 7.0   | 7.1   | 8.9   | 8.0   | 7.3   | 6.9   | 7.1   | 5.9   | 5.0   | 4.2   |
|                                                                        |             | MCDBR  | 6.6   | 6.4   | 10.8  | 12.9  | 13.7  | 10.7  | 8.5   | 8.2   | 9.1   | 9.7   | 7.8   | 6.0   |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.288 | 0.293 | 0.319 | 0.354 | 0.385 | 0.395 | 0.391 | 0.385 | 0.372 | 0.342 | 0.307 | 0.282 |
|                                                                        | taud        |        | 2.433 | 2.377 | 2.391 | 2.357 | 2.349 | 2.358 | 2.392 | 2.408 | 2.427 | 2.485 | 2.520 | 2.489 |
|                                                                        | Ebn at Noon |        | 831   | 894   | 915   | 906   | 884   | 874   | 873   | 865   | 847   | 827   | 805   | 804   |
|                                                                        | Edh at Noon |        | 71    | 90    | 102   | 116   | 121   | 120   | 115   | 109   | 99    | 81    | 65    | 61    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.18  | 2.10  | 3.55  | 4.60  | 5.62  | 6.11  | 6.16  | 5.23  | 4.03  | 2.35  | 1.27  | 0.87  |
|                                                                        | RadStd      |        | 0.12  | 0.22  | 0.37  | 0.46  | 0.48  | 0.44  | 0.25  | 0.33  | 0.35  | 0.28  | 0.18  | 0.09  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (45 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page